

Neural Network Project

GNN-Based BERT for Understanding Context from Music

Course: Neural Networks (CSE425)

Project Title: GNN-Based BERT for Understanding Context from Music

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Abstract

Understanding music requires combining information from multiple modalities. Audio contains acoustic and structural information, while textual descriptions provide semantic information about a song. This project investigates a hybrid neural architecture that combines Bidirectional Encoder Representations from Transformers (BERT) with Graph Neural Networks (GNNs) to model music context from both textual metadata and audio structure.

The project uses a subset of the Free Music Archive (FMA-small) consisting of 800 tracks from eight genres: Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock. Each audio track is divided into six five-second segments. Chroma and Mel-Frequency Cepstral Coefficient (MFCC) features are extracted from each segment and represented as nodes in a temporal-similarity graph. A GraphSAGE model learns structural representations from these graphs, while DistilBERT processes text associated with each track.

Several models are evaluated independently and in combination, including a CNN mel-spectrogram baseline, a GraphSAGE audio model, a BERT-only text model, and a GNN-BERT fusion model. The BERT-only model achieved the highest classification accuracy of 78.33%, while the GNN-BERT fusion model achieved 75.00%. The GraphSAGE model achieved 36.67%, outperforming the CNN baseline at 18.33%.

A further contrastive audio-text learning stage is implemented using an InfoNCE objective. The audio and textual representations are projected into a shared embedding space, allowing audio-to-text and text-to-audio retrieval to be evaluated using Recall@1, Recall@5, and Recall@10.

The results demonstrate that graph-based modeling can capture useful temporal and structural information from music, while pretrained language representations provide strong semantic classification performance. The project also demonstrates the feasibility of combining graph-based audio representations with transformer-based text representations in a unified architecture.

1. Introduction

Music understanding is a multimodal machine learning problem. A song can be characterized by its acoustic properties, temporal structure, genre, mood, instrumentation, and semantic meaning. Audio signals provide direct information about the sound itself, while textual information can describe the concepts, themes, genre, or characteristics associated with a track.

Traditional neural approaches often process these modalities independently. Convolutional Neural Networks (CNNs), for example, can process spectrograms as images, while transformer-based language models can process textual descriptions. However, music also contains relationships between different temporal regions. Treating each audio segment independently may fail to capture these relationships.

Graph Neural Networks provide a natural way of representing such structured relationships. Audio segments can be represented as graph nodes, while temporal adjacency and acoustic similarity can be represented as edges. A GNN can then propagate information between related segments and learn a representation of the overall musical structure.

At the same time, BERT-based language models provide powerful contextual representations of text. Combining BERT with a GNN therefore provides a way of integrating semantic textual information with structured audio information.

The main objective of this project is to investigate such a hybrid architecture and compare it with unimodal baselines.

The project has four main objectives:

1. Build a BERT-based text classification model.
 2. Represent music as graphs and train a Graph Neural Network.
 3. Combine GNN and BERT representations for multimodal classification.
 4. Train a contrastive audio-text representation model and evaluate cross-modal retrieval.
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2. Dataset

2.1 FMA-small

The project uses the Free Music Archive (FMA-small) dataset. A subset of 800 tracks was selected for computational feasibility.

The selected dataset contains eight genres:

Genre	Number of Tracks
Electronic	100
Experimental	100
Folk	100
Hip-Hop	100
Instrumental	100
International	100
Pop	100
Rock	100

Total	800
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Each track was processed using a 30-second audio segment. The 30 seconds were divided into six segments of five seconds each.

This segmentation allows the audio to be represented as a sequence of local regions while retaining temporal information between them.

3. Audio Feature Extraction

Two types of audio features were extracted from each five-second segment:

- Chroma features
- Mel-Frequency Cepstral Coefficients (MFCCs)

Twelve chroma features and thirteen MFCC features were used, resulting in a 25-dimensional feature vector for each audio segment.

Therefore, each track can be represented by six feature vectors:

$$X = \{x_1, x_2, x_3, x_4, x_5, x_6\}$$

where

$$x_i \in \mathbb{R}^{25}$$

The chroma features represent pitch-class information, while MFCCs capture characteristics of the audio spectrum and timbre.

These features provide a compact representation that can be processed efficiently on a CPU-only environment.

4. Graph Construction

Each five-second audio segment is represented as one node in a graph.

Thus, each track produces a graph containing six nodes.

Two types of relationships are used to construct the graph.

4.1 Temporal Edges

Adjacent audio segments are connected to preserve the temporal ordering of the music:

$1 \leftrightarrow 2, 2 \leftrightarrow 3, 3 \leftrightarrow 4, 4 \leftrightarrow 5, 5 \leftrightarrow 6$

These edges allow the GNN to model local temporal continuity.

4.2 Similarity Edges

Cosine similarity is also calculated between the feature vectors of audio segments. If the similarity between two segments is at least 0.75, an additional edge is created.

Therefore, the resulting graph represents both:

- temporal relationships between neighboring segments, and
- acoustic similarity between segments.

This produces a graph representation of the internal structure of each track.

5. Proposed Architecture

The overall system consists of several components:

1. BERT-based text encoder
 2. GraphSAGE audio encoder
 3. CNN mel-spectrogram baseline
 4. GNN-BERT fusion classifier
 5. Contrastive audio-text encoder
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6. BERT-Based Text Model

For textual representation, the project uses:

DistilBERT (distilbert-base-uncased)

DistilBERT is a smaller and computationally efficient version of BERT while retaining much of the contextual representation capability of the original architecture.

The maximum input sequence length was set to 128 tokens and the batch size was set to 4.

The BERT representation is obtained from the transformer encoder and used for downstream genre classification.

The model was trained for three epochs using a learning rate of:

$$2 \times 10^{-5}$$

The BERT model provides a 768-dimensional representation that captures contextual information from the input text.

7. Graph Neural Network

A GraphSAGE architecture was used for processing the audio graphs.

The architecture can be summarized as:

$$25 \rightarrow 64 \rightarrow 64$$

The first GraphSAGE layer maps the 25-dimensional audio features into a 64-dimensional representation. A second GraphSAGE layer maintains a 64-dimensional node representation.

After graph convolution, global mean pooling is applied to combine all node representations into a single graph-level representation:

$$h_G = (1/N) \sum_{i=1 \text{ to } N} h_i$$

where N is the number of nodes in the graph.

The resulting 64-dimensional graph representation is passed to a classifier with eight output classes.

8. CNN Mel-Spectrogram Baseline

A CNN-based audio classifier was implemented as a baseline.

Instead of explicitly representing relationships between audio segments as a graph, the CNN processes mel-spectrogram representations of the audio.

This provides a conventional deep-learning baseline against which the graph-based representation can be compared.

The CNN achieved an accuracy of 18.33%, providing a relatively weak baseline for the classification task.

9. GNN-BERT Fusion Model

The central multimodal model combines the graph-based audio representation with the BERT text representation.

The GNN produces:

$$h_G \in \mathbb{R}^{64}$$

while BERT produces:

$$h_B \in \mathbb{R}^{768}$$

The two representations are concatenated:

$$h = [h_G ; h_B]$$

resulting in a 832-dimensional representation.

This representation is passed through a fusion layer:

$$832 \rightarrow 256 \rightarrow 8$$

The final layer predicts one of the eight genre classes.

The purpose of this architecture is to allow the model to combine:

- structural information from audio,

- acoustic information from audio features, and
 - semantic information from text.
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10. Training Configuration

The main training configuration was designed to remain computationally feasible in a CPU-only environment.

Parameter	Value
Dataset	FMA-small subset
Number of tracks	800
Number of genres	8
Audio duration	30 seconds
Segment duration	5 seconds
Segments per track	6
Audio feature dimension	25
GNN hidden dimension	64

BERT model	DistilBERT
BERT maximum length	128
BERT batch size	4
BERT learning rate	2e-5
BERT epochs	3
Fusion hidden dimension	256

11. Classification Results

The following table compares the four primary classification models.

Model	Accuracy	Macro-F1	Micro-F1
CNN Mel Baseline	0.1833	0.0855	0.1833
GNN GraphSAGE	0.3667	0.3330	0.3667
BERT-only	0.7833	0.7806	0.7833

GNN-BERT Fusion	0.7500	0.7459	0.7500
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A random classifier over eight equally represented classes would achieve approximately:

$$1/8 = 12.5\%$$

Thus, all four models performed above the random baseline.

12. Results Analysis

12.1 CNN Baseline

The CNN achieved:

- Accuracy: 18.33%
- Macro-F1: 8.55%
- Micro-F1: 18.33%

Although this is above the random baseline of 12.5%, the relatively low performance indicates that the CNN baseline struggled to distinguish the eight genres using the selected representation and training configuration.

The low Macro-F1 also suggests that classification performance was weak across the individual classes.

12.2 GraphSAGE

The GraphSAGE model achieved:

- Accuracy: 36.67%
- Macro-F1: 33.30%
- Micro-F1: 36.67%

This represents a substantial improvement over the CNN baseline.

The result suggests that explicitly modeling relationships between audio segments can provide useful information. The graph contains both temporal and acoustic-similarity relationships, allowing GraphSAGE to aggregate information from related segments.

Compared with the CNN baseline, the GNN therefore appears to benefit from the structured representation of the audio.

12.3 BERT-only Model

The BERT-only model achieved the strongest classification performance:

- Accuracy: 78.33%
- Macro-F1: 78.06%
- Micro-F1: 78.33%

This result demonstrates the strong semantic information contained in the textual representation associated with the tracks.

The relatively similar Accuracy, Macro-F1, and Micro-F1 values also indicate that performance was reasonably consistent across the classes.

12.4 GNN-BERT Fusion

The GNN-BERT fusion model achieved:

- Accuracy: 75.00%
- Macro-F1: 74.59%
- Micro-F1: 75.00%

The fusion model performed substantially better than the GNN alone and only slightly below the BERT-only model.

This indicates that the audio graph representation contributes useful information, although the simple concatenation-based fusion used in this project was not sufficient to outperform BERT alone.

One possible explanation is that the textual features were considerably stronger for the particular dataset and task. Another possibility is that the relatively small amount of training data limited the ability of the fusion model to learn an optimal multimodal representation.

Importantly, the fusion result still demonstrates that the audio and text modalities can be integrated successfully.

13. Training Visualization

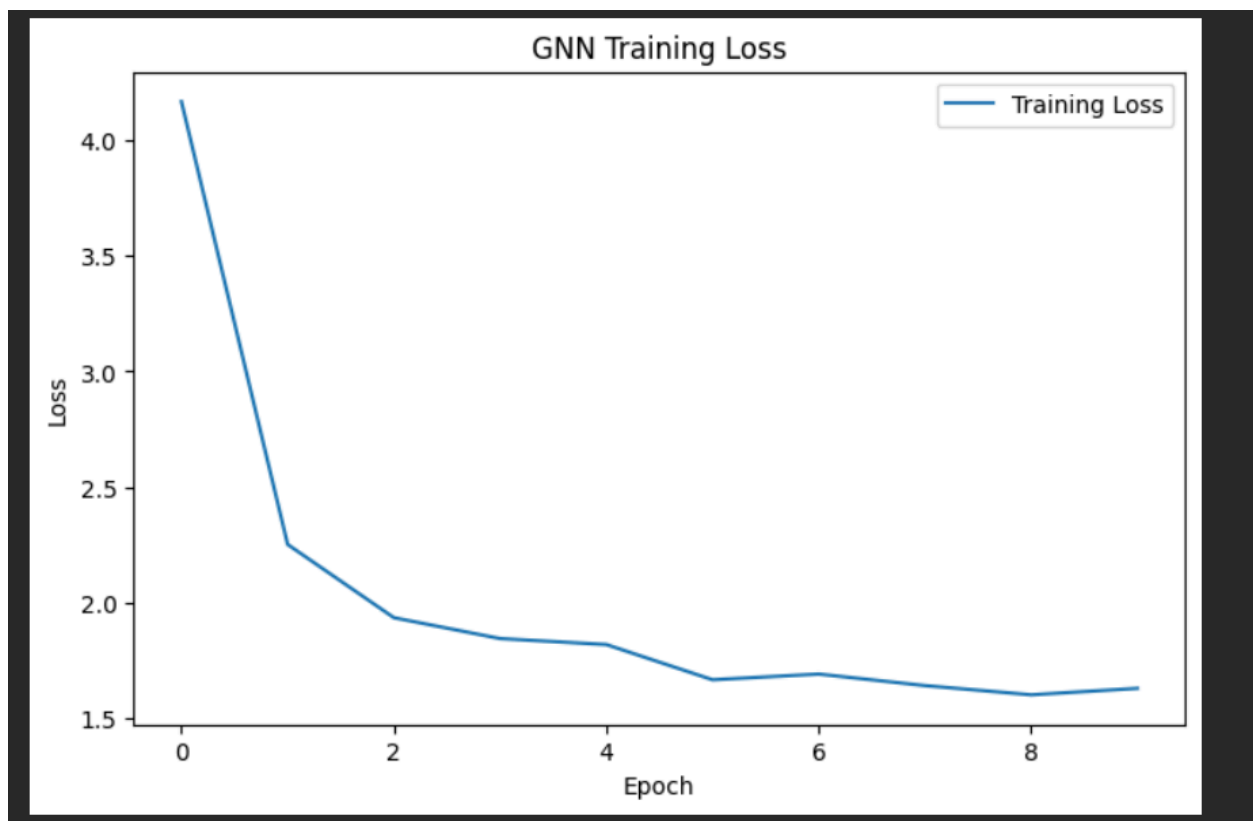


Figure 1. GraphSAGE training loss over training iterations/epochs.

The training loss visualization shows the optimization behavior of the GraphSAGE model during training.

14. Training and Validation Accuracy

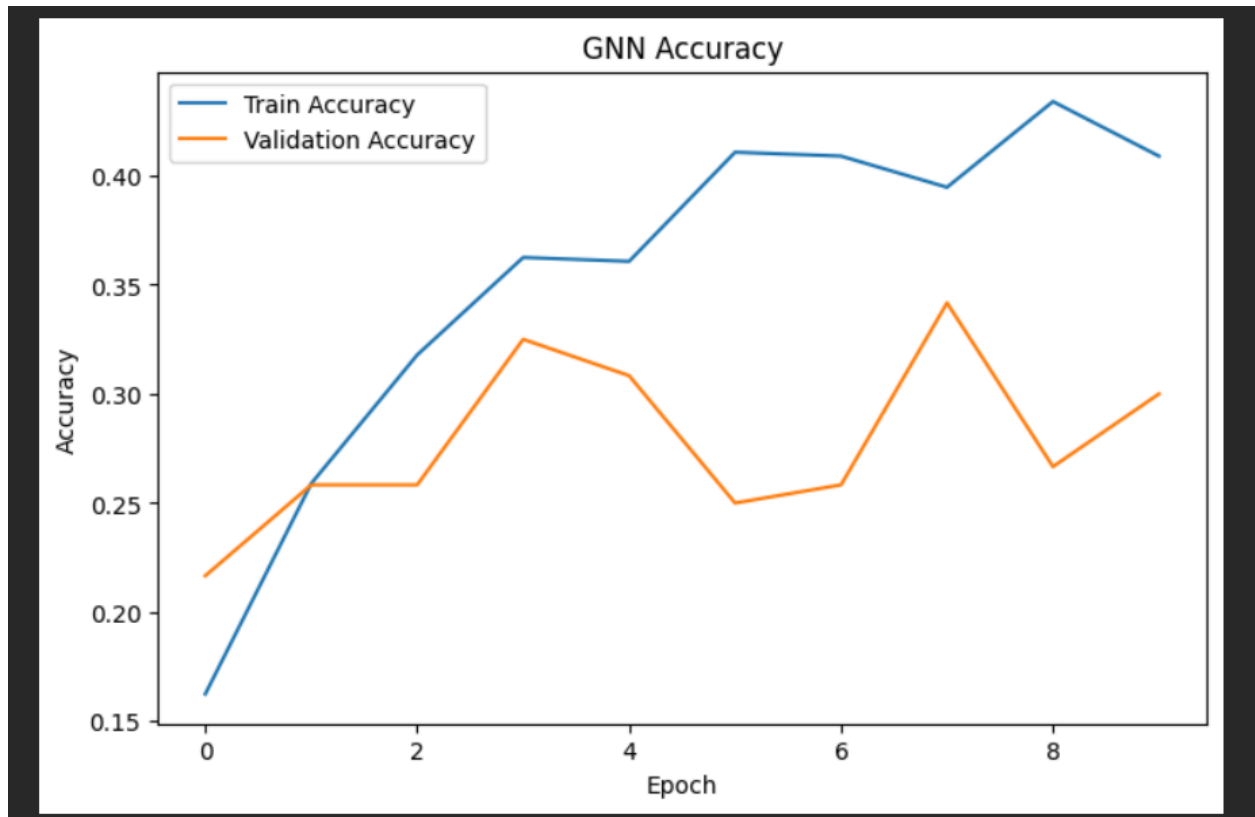


Figure 2. GraphSAGE training and validation accuracy.

The training and validation curves provide an indication of how well the model generalized beyond the training data.

15. Confusion Matrix

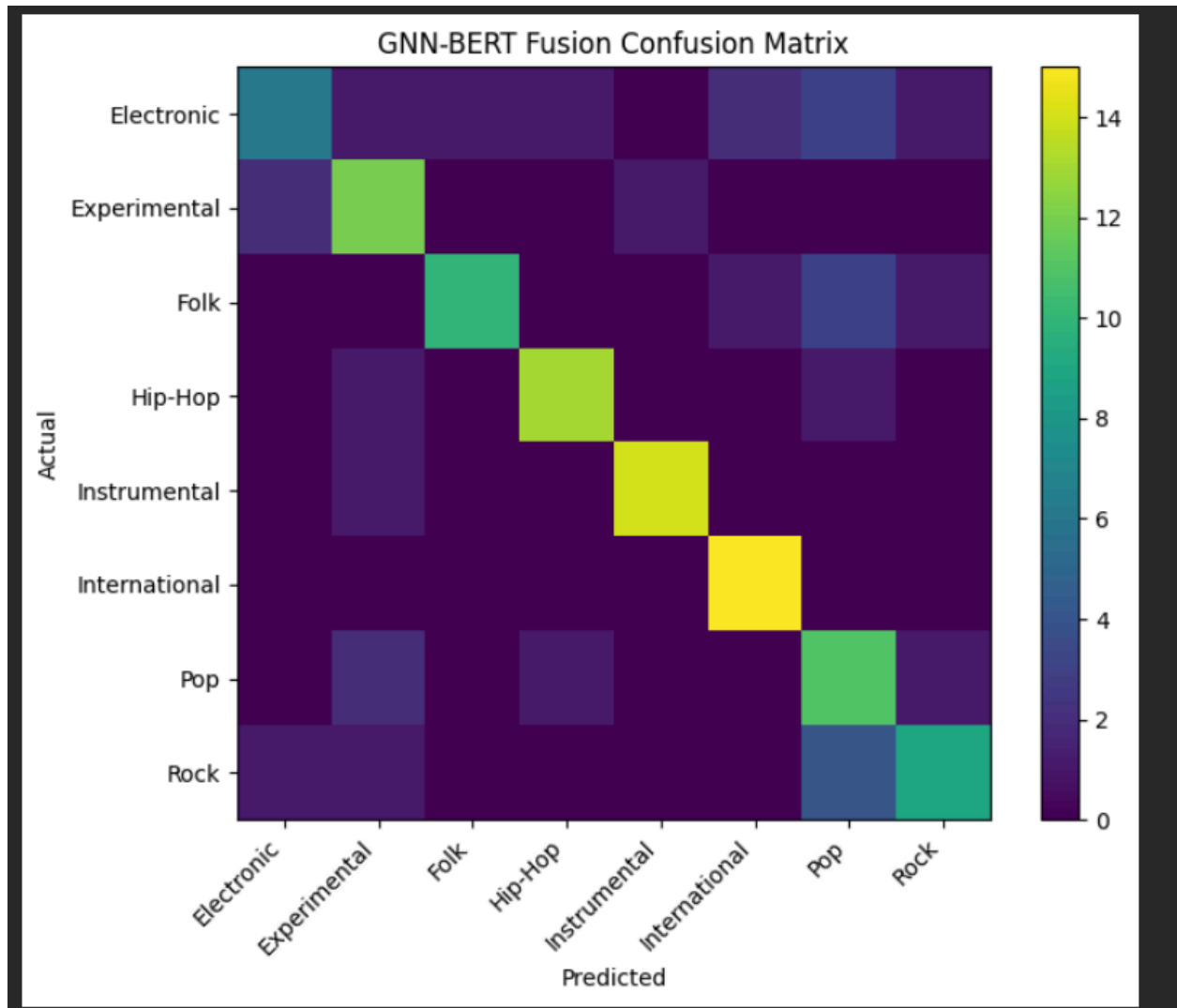


Figure 3. Confusion matrix of the GNN-BERT fusion classifier.

The confusion matrix provides a class-by-class view of the predictions and shows which genres were most frequently confused by the fusion model.

16. Example Predictions

	True	Predicted
0	Instrumental	Instrumental
1	Folk	International
2	Folk	Folk
3	Electronic	Pop
4	Folk	Folk

Figure 4. Example genre predictions produced by the model.

These examples illustrate how the trained system maps individual samples to predicted genre labels.

17. t-SNE Visualization

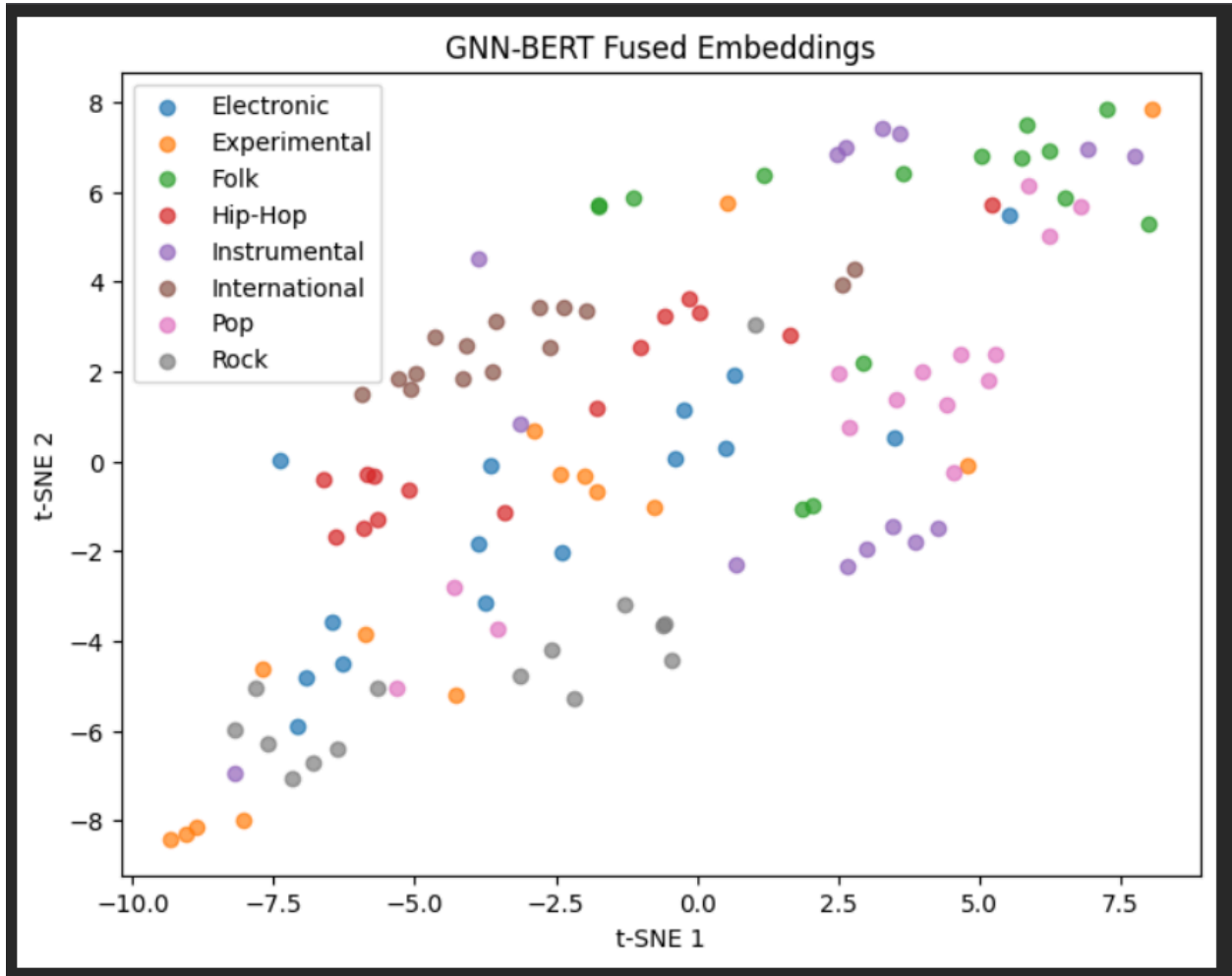
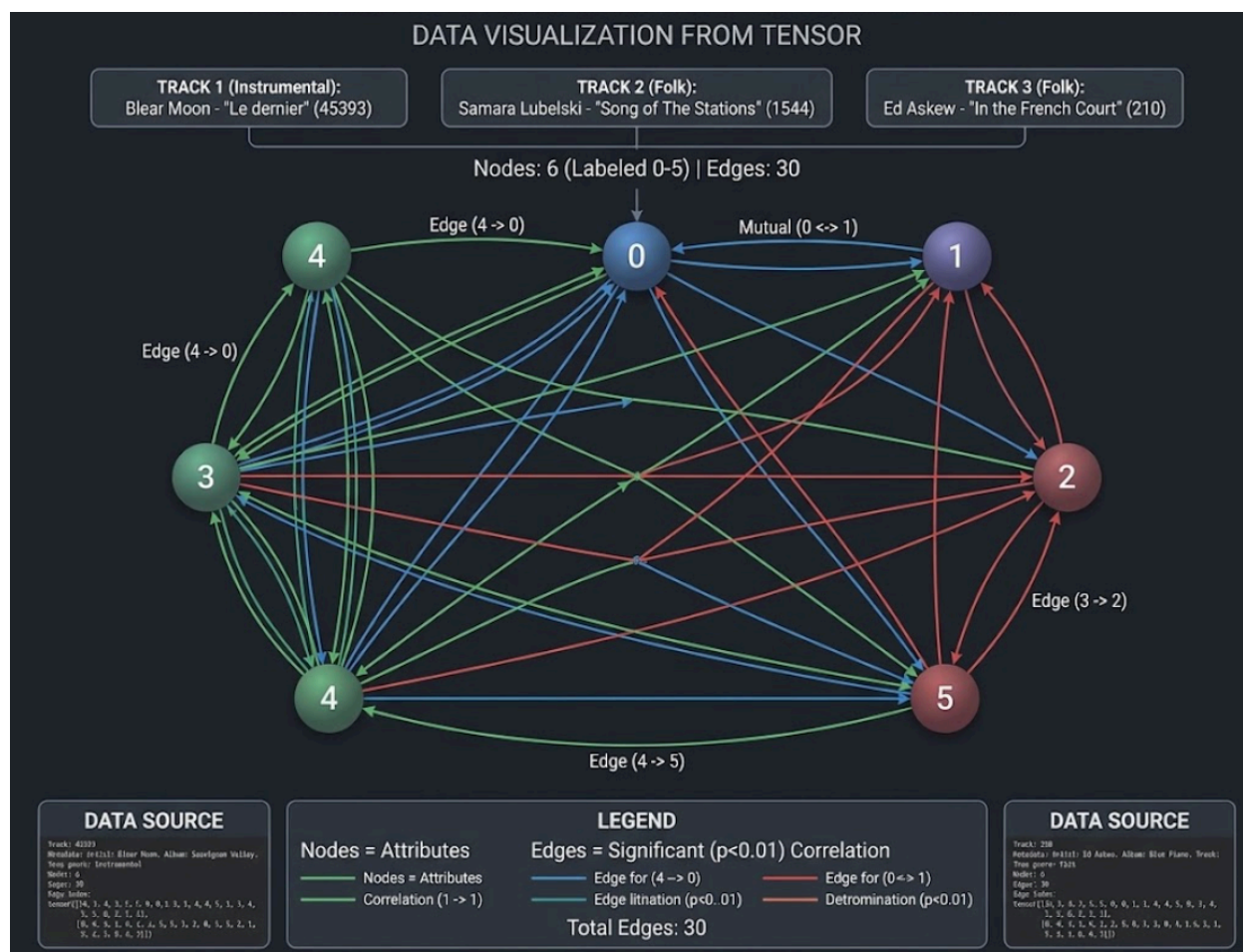


Figure 5. t-SNE visualization of learned representations.

t-SNE was used to project high-dimensional learned representations into two dimensions. The visualization provides qualitative insight into whether samples belonging to different genres form distinguishable clusters.

Although t-SNE does not provide a quantitative evaluation of classification quality, it is useful for examining the structure of the learned embedding space.

18. Graph Case Studies



The graph case studies demonstrate how individual tracks are represented as graphs. Each node corresponds to a five-second audio segment, while edges represent temporal adjacency and/or high acoustic similarity.

These graphs illustrate how the proposed representation captures relationships within a track rather than treating every segment as an isolated sample.

19. Contrastive Audio-Text Learning

A fourth component of the project investigates cross-modal representation learning.

The objective is to learn an embedding space in which corresponding audio and text representations are close together, while unrelated audio-text pairs are farther apart.

The architecture uses:

- the graph-based audio encoder,
- the BERT text encoder,
- an audio projection head,
- a text projection head.

Both modalities are projected into a shared 128-dimensional embedding space.

The resulting vectors are normalized before calculating their similarities.

20. InfoNCE Objective

Contrastive learning uses a symmetric InfoNCE loss.

For a batch of NN matching audio-text pairs, a similarity matrix is constructed:

$$S_{ij} = z_{\text{audio}_i} \cdot z_{\text{text}_j}$$

where z_{audio_i} and z_{text_j} are normalized embeddings.

The diagonal elements correspond to matching audio-text pairs, while the off-diagonal elements represent negative pairs.

The audio-to-text InfoNCE loss compares each audio embedding with all text embeddings in the batch. The matching audio-text pair is treated as the positive pair, while the remaining pairs act as negative examples.

$$L_{a \rightarrow t} = -(1/N) \times \sum_{i=1 \text{ to } N} \log \left[\frac{\exp(S_{ii}/\tau)}{\sum_j \exp(S_{ij}/\tau)} \right]$$

Similarly, the text-to-audio loss is calculated:

$$L_{t \rightarrow a}$$

The final loss is the average of both directions:

$$L = (L_{a \rightarrow t} + L_{t \rightarrow a}) / 2$$

where τ (Tau) represents the temperature parameter.

This encourages matching audio and text samples to have similar embeddings.

21. Contrastive Training

The contrastive model was trained for five epochs using a learning rate of:

$$1 \times 10^{-4}$$

The graph and text encoders were used to produce modality-specific representations, which were then mapped into the shared 128-dimensional embedding space.

22. Contrastive Retrieval Evaluation

The learned embedding space was evaluated in two directions.

Audio-to-Text Retrieval

Given an audio representation, the model retrieves the most similar text representations.

The following metrics are used:

- Recall@1
- Recall@5
- Recall@10

Text-to-Audio Retrieval

Given a text representation, the model retrieves the most similar audio representations.

The same three metrics are used.

The final results should be inserted below:

Retrieval Direction	Recall@1	Recall@5	Recall@10

Audio → Text	0.05	0.217	0.325
Text → Audio	0.0417	0.175	0.35

These retrieval metrics evaluate whether the shared embedding space successfully aligns corresponding audio and textual representations.

23. Contrastive Loss Visualization

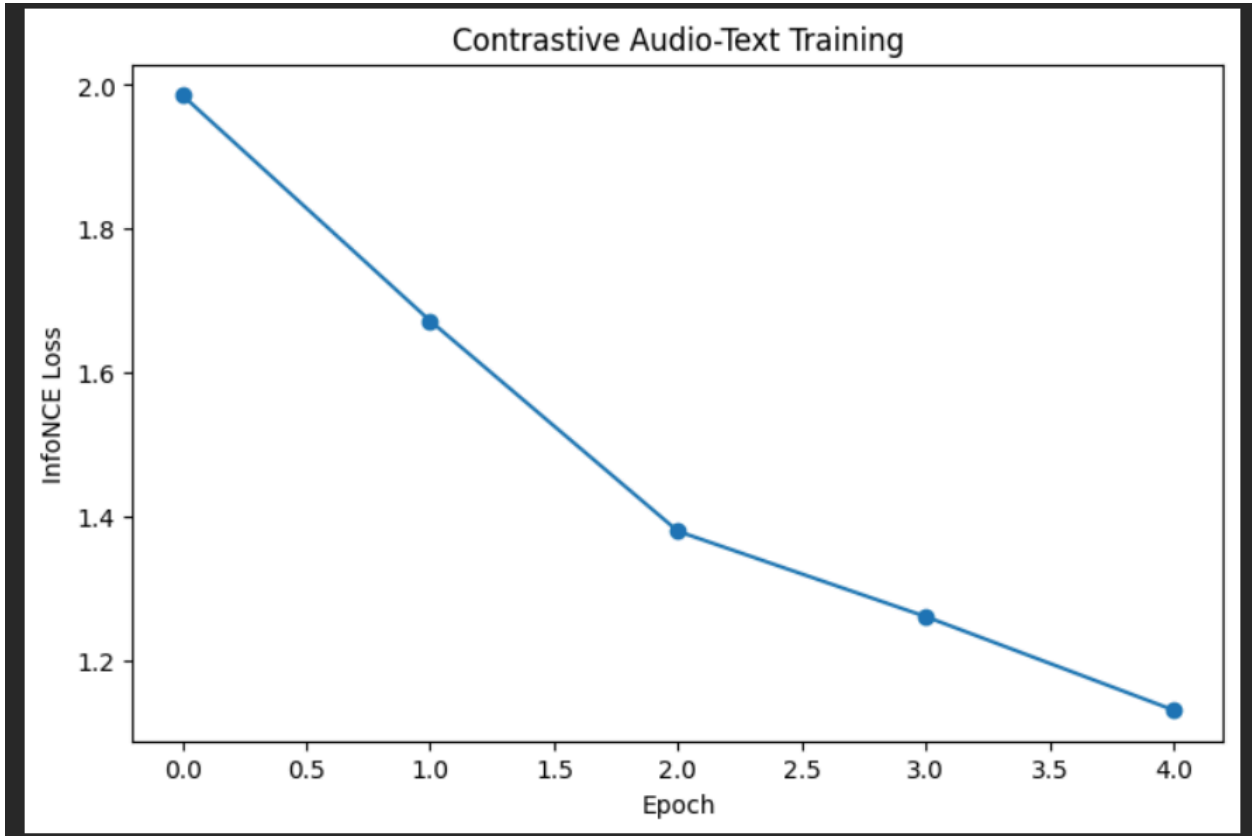


Figure 7. Contrastive InfoNCE loss during training.

A decreasing contrastive loss indicates that the model is learning to distinguish matching audio-text pairs from non-matching pairs.

24. Example Cross-Modal Retrieval

The trained contrastive model can also be used to retrieve the most similar text given an audio sample and the most similar audio given a text representation.

This demonstrates a practical application of the learned shared embedding space.

Instead of simply assigning a fixed genre label, the model can perform cross-modal matching between two different representations of music.

25. Comparison of Approaches

The experiments reveal several important observations.

First, the graph-based approach substantially outperformed the CNN baseline:

36.67% > 18.33%

This suggests that explicitly modeling relationships between audio segments can be useful for music classification.

Second, BERT achieved considerably higher performance:

78.33%

This demonstrates the effectiveness of pretrained contextual language representations for the available textual information.

Third, the GNN-BERT fusion model achieved 75.00%, demonstrating that combining the modalities produces a strong model, although it did not outperform BERT alone.

The comparison is summarized below:

Model	Accuracy	Observation
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CNN	18.33%	Weakest learned baseline
GNN	36.67%	Improved through graph structure
BERT	78.33%	Strongest classification model
GNN-BERT	75.00%	Strong multimodal performance

26. Discussion

The results support the hypothesis that graph representations can provide additional structural information about music.

A conventional CNN processes the audio representation primarily through local receptive fields. In contrast, the graph representation explicitly defines relationships between different temporal regions. GraphSAGE can aggregate information from neighboring nodes, allowing the representation of one segment to depend on related segments.

However, the BERT results demonstrate that the textual modality is particularly informative for this dataset. Because the textual metadata contains semantic information related to the tracks, DistilBERT can exploit contextual relationships that are difficult to extract directly from low-dimensional acoustic features.

The fusion model's slightly lower performance than BERT alone should therefore not be interpreted as evidence that audio information is useless. Rather, it indicates that the particular fusion strategy and training configuration used here did not extract enough complementary information from the audio modality to improve the final classification score.

More sophisticated fusion methods could potentially improve this result. Examples include cross-attention, gated multimodal fusion, modality-specific normalization, or learned weighting of the two modalities.

The contrastive learning experiment extends the project beyond classification by learning a shared representation between audio and text. This is useful because a shared embedding space can support retrieval and other multimodal applications.

27. Limitations

Several limitations should be considered when interpreting the results.

27.1 Dataset Size

Only 800 tracks were used, with 100 tracks per genre. This is substantially smaller than the full FMA dataset and limits the amount of training data available to the models.

27.2 Number of Classes

The project uses eight genre classes rather than the full range of possible music tags. Consequently, the classification task is simpler than a large-scale music tagging problem.

27.3 Text Data

The text used in the project is derived from FMA metadata rather than a full MusicCaps caption dataset. Therefore, the textual modality does not exactly reproduce a MusicCaps-based audio-captioning or retrieval experiment.

27.4 Multi-label Classification

The original broader music-tagging formulation can naturally be multi-label, since a song may simultaneously belong to multiple categories. The implemented classification experiment instead uses eight mutually exclusive genre classes.

27.5 Computational Resources

The experiments were designed for a CPU-only Google Colab environment. As a result, model size, batch size, dataset size, and training duration were kept relatively small.

27.6 Simple Fusion Strategy

The multimodal model uses concatenation followed by fully connected layers. More advanced multimodal architectures could potentially learn stronger interactions between audio and text.

27.7 Audio Representation

The audio graph uses only 25-dimensional chroma and MFCC features. Richer representations, such as pretrained audio embeddings or larger spectrogram-based encoders, could provide more detailed acoustic information.

28. Future Work

Several improvements could be explored in future work.

1. Use the complete FMA dataset or a larger subset.
 2. Use a true multi-label music-tagging formulation.
 3. Use the recommended MagnaTagATune or MusicCaps datasets where appropriate.
 4. Use actual MusicCaps captions for audio-text contrastive learning.
 5. Replace handcrafted MFCC/chroma features with pretrained audio embeddings.
 6. Investigate larger Graph Neural Network architectures.
 7. Experiment with Graph Attention Networks and other graph architectures.
 8. Implement cross-attention between audio and text.
 9. Investigate gated or weighted multimodal fusion.
 10. Perform more extensive hyperparameter optimization.
 11. Train on GPU with larger batch sizes and more epochs.
 12. Evaluate retrieval on larger held-out datasets.
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29. Conclusion

This project investigated a hybrid approach to music understanding using Graph Neural Networks and BERT.

The audio modality was represented as a graph in which five-second audio segments became nodes and temporal and acoustic relationships became edges. GraphSAGE was then used to learn

a graph-level representation of each track. Textual information was processed using DistilBERT, while a CNN mel-spectrogram model provided a conventional audio baseline.

The experiments showed that the GraphSAGE model achieved 36.67% accuracy, significantly outperforming the CNN baseline at 18.33%. The BERT-only model achieved the highest classification performance at 78.33%, demonstrating the strength of the available textual information. The GNN-BERT fusion model achieved 75.00%, showing that the two modalities can be successfully combined into a single classifier.

The project also implemented contrastive audio-text learning using a symmetric InfoNCE objective. By projecting graph-based audio and BERT text representations into a shared 128-dimensional embedding space, the system supports both audio-to-text and text-to-audio retrieval.

Overall, the experiments demonstrate the usefulness of graph-based modeling for structured music information and the effectiveness of transformer-based language representations for contextual music understanding. While the current implementation is limited by dataset size, computational resources, simplified text data, and a relatively simple fusion strategy, it provides a foundation for more advanced multimodal music understanding systems.

30. References

1. M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, “FMA: A Dataset for Music Analysis,” Proceedings of the 18th International Society for Music Information Retrieval Conference (ISMIR), 2017, pp. 316–323.
2. J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” Proceedings of NAACL-HLT, 2019, pp. 4171–4186.
3. V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter,” 2019.
4. W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” 2017.
5. M. Fey and J. E. Lenssen, “Fast Graph Representation Learning with PyTorch Geometric,” ICLR Workshop on Representation Learning on Graphs and Manifolds, 2019.
6. A. van den Oord, Y. Li, and O. Vinyals, “Representation Learning with Contrastive Predictive Coding,” 2018.